

Plan: GömbSoma-Ricci-Stim Cluttered MNIST falsification battery

2026-05-15

1 Scope

Train `RicciStimDetector` on Cluttered MNIST and compare against the published HyMeYOLO baselines. The target, from the Ricci-Stim plan §10:

- HyMeYOLO +`kcycle` (broken): 0.204 mAP50 — beat strongly.
- HyMeYOLO +`ricci+kcycle`: 0.235 mAP50 — beat strongly.
- HyMeYOLO +`ricci-mod` (positive control): 0.723 mAP50 — match or beat.
- Phase 3-V rolling-shutter GömbSoma: beat.

2 Ablation matrix

Phase 10 exposes the full grid:

config	α	β	SDRF	what it tests
A	0	0	off	bare backbone (Phase 7 baseline)
B	0.1	0	off	Hodge smoothing alone
C	0	0.1	off	Ricci correction alone
D	0.1	0.1	off	full Bochner without graph rewiring
E	0.1	0.1	on	headline config

3 Affected files

- `signedkan_wip/src/hymeko_gomb/soma/vision/ricci_stim_train.py` — training infrastructure: anchor assignment, loss, train loop, mAP50 evaluator.
- `signedkan_wip/experiments/run_ricci_stim_cluttered_mnist_2026_05_15.sh` — orchestration script.
- `signedkan_wip/tests/test_gomb_soma_vision_ricci_stim_train.py` — unit tests for the assignment + loss + smoke train.

4 Performance budget

- Per-config wall: ≤ 30 min on the RTX 2070 SUPER (5000 images $\times$ 20 epochs $\times$ 5 seeds is the HyMeYOLO baseline budget).
- Total: 5 configs $\times$ 5 seeds $\approx$ 2.5–3 hours.
- Peak RSS: ≤ 14 GB cgroups MemoryMax 16 GB.

5 Test strategy

- Unit: anchor-target IoU assignment correctness on small synthetic.
- Unit: loss computation on hand-built positives + negatives.
- Integration smoke: 50 images $\times$ 2 epochs $\times$ 1 seed (≤ 1 min wall) — pipeline runs, training loss decreases.
- Full battery: 5 configs $\times$ 5 seeds (queued; multi-hour).

6 CORE.YAML items touched

None.

7 Decision tree

- Config E reaches $\text{mAP}_{50} \geq 0.72 \Rightarrow$ paper-grade YOLO-class result; write up as headline contribution.
- Config E reaches $\text{mAP}_{50} \in [0.5, 0.72) \Rightarrow$ partial confirmation; ablate to isolate which component contributes.
- Config E reaches $\text{mAP}_{50} \in [0.235, 0.5) \Rightarrow$ beats the broken HyMeYOLO baseline; partial win; document.
- Config E reaches $\text{mAP}_{50} < 0.235 \Rightarrow$ stimulus-driven hypothesis falsified; document as negative result.

8 Risk anticipation

- *Anchor-target assignment is non-trivial.* Many detection papers spend pages on this. Use simple max-IoU assignment with positive threshold 0.3, negative threshold 0.1.
- *NMS implementation.* Standard greedy NMS at IoU 0.5; no learned variants.
- *Multi-scale anchor scales.* Cluttered MNIST has 28-px digits on a 64-px canvas. Use `patch_size_initial = 32`, `max_depth = 3` to give anchors from 4 to 32 px.
- *Per-image variable anchor count.* Loss is computed per-image then averaged across batch. Standard pattern.
- *Empty-plan-dir hygiene.* If shelved, delete the plan dir.

9 Bottom line

A real experiment on a real dataset against published baselines. Architectural infrastructure is complete (phases 1–10); this is the empirical validation. Either we beat YOLO at its own game, or we have a clean negative result with a known architecture and a published baseline reference.